

Research Profile

Postdoctoral researcher in systems and control, working at the intersection of model predictive control and learning-based control. My research focuses on the design of control methods for autonomous and networked systems with formal guarantees on stability, safety, and constraint satisfaction, with particular interest in distributed multi-agent systems, neural controllers, and online adaptation.

Research areas: model predictive control; learning-based control; distributed and multi-agent control; neural controllers with formal guarantees; robust and safe control.

Academic and Professional Experience

Apr 2023 – present	Postdoctoral Researcher , DECODE group, École polytechnique fédérale de Lausanne (EPFL), Lausanne, Switzerland. Research on learning-based and distributed control, MPC, and neural controllers with stability and safety guarantees.
Mar 2022 – Sep 2022	Visiting PhD Researcher , Institute for Dynamic Systems and Control, ETH Zurich, Zurich, Switzerland. Host: Prof. Melanie N. Zeilinger. Research on distributed control of multi-agent systems and MPC for robotic systems.
Sep 2019 – Nov 2019	Research Assistant , Politecnico di Milano, Milan, Italy. Modelling and control development for safety-critical autonomous systems in MATLAB and ROS.
Sep 2014 – Sep 2016	R&D Mechanical Engineer , Renova s.r.l., Milan, Italy. Mechanical design, prototyping, technical documentation, and development of custom web-tension control solutions.

Education

Nov 2019 – Mar 2023	PhD in Information Technology, cum laude , Politecnico di Milano, Milan, Italy. Thesis: <i>Model Predictive Control for Constrained Navigation of Autonomous Vehicles</i> . Supervisor: Prof. Lorenzo Fagiano.
Sep 2016 – Apr 2019	MSc in Automation and Control Engineering, cum laude , Politecnico di Milano, Milan, Italy. Thesis on learning-based hybrid predictive control of a large-scale cooling system using recurrent neural-network models. Supervisor: Prof. Riccardo Scattolini.
2010 – 2014	BSc in Mechanical Engineering , Politecnico di Milano, Milan, Italy.

Selected Research Contributions

- **Predictive control for safe autonomous systems.** Developed model predictive control methods for constrained autonomous systems, with particular emphasis on reducing conservatism while preserving recursive feasibility, stability, and constraint satisfaction. Contributions include multi-trajectory MPC, robust MPC with multi-step predictors, and predictive stability filters.
- **Distributed MPC and multi-agent control.** Developed distributed predictive-control methods for multi-agent systems with limited communication, time-varying interaction graphs, connectivity requirements, and plug-and-play operation. Recent work focuses on contract-based formulations that preserve global properties through local constraints and neighbor-to-neighbor communication.
- **Stability-preserving learning-based control.** Developed learning-based control architectures in which neural controllers can improve closed-loop performance while stability and safety properties are preserved by design. Contributions include distributed neural closed-loop maps, constrained performance boosting, and stability-preserving online controller adaptation.

Awards and Distinctions

- 2024** **Finalist**, IEEE Intelligent Transportation Systems Society Italy Section Best Dissertation Award.
- 2023** **PoliMi SpringerBriefs Award**, annual distinction for outstanding PhD theses at Politecnico di Milano.

Teaching

- 2025** **Lecturer and Organizer**, *Neural Networks for Optimal Control*, EECI International Graduate School on Control, EPFL.
- 2024** **Lecturer and Organizer**, *Distributed Model Predictive Control*, ETH Zurich.
- 2020 – 2022** **Teaching Assistant**, *Fondamenti di Automatica*, Politecnico di Milano, four semesters.
- 2019** **Instructor**, finite-element analysis computer laboratory, Fondazione Istituto Tecnico Superiore per le Nuove Tecnologie della Vita, Bergamo, Italy.

Publications

Peer-reviewed journal articles

- [J1] G. Giacomelli, D. Saccani, S. Weiland, G. Ferrari-Trecate, and V. Breschi, “Constrained performance boosting control for nonlinear systems,” *Engineering Applications of Artificial Intelligence*, vol. 182, p. 115842, 2026. DOI: [10.1016/j.engappai.2026.115842](https://doi.org/10.1016/j.engappai.2026.115842)
- [J2] D. Saccani, L. Cecchin, and L. Fagiano, “Multitrajectory model predictive control for safe UAV navigation in an unknown environment,” *IEEE Transactions on Control Systems Technology*, vol. 31, no. 5, pp. 1982–1997, 2023. DOI: [10.1109/TCST.2022.3216989](https://doi.org/10.1109/TCST.2022.3216989)
- [J3] D. Saccani, G. Ferrari-Trecate, M. N. Zeilinger, and J. Köhler, “Homothetic tube model predictive control with multi-step predictors,” *IEEE Control Systems Letters*, vol. 7, pp. 3561–3566, 2023. DOI: [10.1109/LCSYS.2023.3336261](https://doi.org/10.1109/LCSYS.2023.3336261) arXiv: [2309.06591](https://arxiv.org/abs/2309.06591).
- [J4] E. Terzi, T. Bonetti, D. Saccani, M. Farina, L. Fagiano, and R. Scattolini, “Learning-based predictive control of the cooling system of a large business centre,” *Control Engineering Practice*, vol. 97, p. 104348, 2020. DOI: [10.1016/j.conengprac.2020.104348](https://doi.org/10.1016/j.conengprac.2020.104348)

Peer-reviewed conference papers

- [C1] D. Ravasio, D. Saccani, M. Farina, and G. Ferrari-Trecate, “Learning stabilising policies for constrained nonlinear systems,” in *2026 European Control Conference (ECC)*, 2026, pp. 491–496. arXiv: [2511.06832](https://arxiv.org/abs/2511.06832).
- [C2] D. Saccani, L. Furieri, and G. Ferrari-Trecate, “Stability-preserving online adaptation of neural closed-loop maps,” in *2026 European Control Conference (ECC)*, 2026, pp. 1917–1922. arXiv: [2603.22469](https://arxiv.org/abs/2603.22469).
- [C3] D. Saccani, H. Shen, L. Furieri, and G. Ferrari-Trecate, “Safety-aware performance boosting for constrained nonlinear systems,” in *2026 European Control Conference (ECC)*, 2026, pp. 593–598. arXiv: [2603.19361](https://arxiv.org/abs/2603.19361).
- [C4] L. Furieri, S. Shenoy, D. Saccani, A. Martin, and G. Ferrari-Trecate, “MAD: A magnitude and direction policy parametrization for stability-constrained reinforcement learning,” in *2025 IEEE 64th Conference on Decision and Control (CDC)*, 2025, pp. 942–947. arXiv: [2504.02565](https://arxiv.org/abs/2504.02565).
- [C5] L. Massai, D. Saccani, L. Furieri, and G. Ferrari-Trecate, “Unconstrained learning of networked nonlinear systems via free parametrization of stable interconnected operators,” in *2024 European Control Conference (ECC)*, 2024, pp. 651–656. DOI: [10.23919/ECC64448.2024.10591242](https://doi.org/10.23919/ECC64448.2024.10591242) arXiv: [2311.13967](https://arxiv.org/abs/2311.13967).
- [C6] D. Saccani, L. Massai, L. Furieri, and G. Ferrari-Trecate, “Optimal distributed control with stability guarantees by training a network of neural closed-loop maps,” in *2024 IEEE 63rd Conference on Decision and Control (CDC)*, 2024, pp. 3776–3781. arXiv: [2404.02820](https://arxiv.org/abs/2404.02820).
- [C7] A. Carron, D. Saccani, L. Fagiano, and M. N. Zeilinger, “Multi-agent distributed model predictive control with connectivity constraint,” *IFAC-PapersOnLine*, vol. 56, no. 2, pp. 3806–3811, 2023. DOI: [10.1016/j.ifacol.2023.10.1310](https://doi.org/10.1016/j.ifacol.2023.10.1310) arXiv: [2303.06957](https://arxiv.org/abs/2303.06957).
- [C8] D. Saccani, L. Fagiano, M. N. Zeilinger, and A. Carron, “Model predictive control for multi-agent systems under limited communication and time-varying network topology,” in *2023 IEEE 62nd Conference on Decision and Control (CDC)*, 2023, pp. 3764–3769. DOI: [10.1109/CDC49753.2023.10383790](https://doi.org/10.1109/CDC49753.2023.10383790) arXiv: [2304.01649](https://arxiv.org/abs/2304.01649).
- [C9] M. Bolognini, D. Saccani, F. Cirillo, and L. Fagiano, “Autonomous navigation of interconnected tethered drones in a partially known environment with obstacles,” in *2022 IEEE 61st Conference on Decision and Control (CDC)*, 2022, pp. 3315–3320. DOI: [10.1109/CDC51059.2022.9992802](https://doi.org/10.1109/CDC51059.2022.9992802)
- [C10] L. Cecchin, D. Saccani, and L. Fagiano, “G-BEAM: Graph-based exploration and mapping for autonomous vehicles,” in *2021 IEEE Conference on Control Technology and Applications (CCTA)*, 2021, pp. 1011–1016. DOI: [10.1109/CCTA48906.2021.9659296](https://doi.org/10.1109/CCTA48906.2021.9659296)

- [C11] D. Saccani and L. Fagiano, “Autonomous UAV navigation in an unknown environment via multi-trajectory model predictive control,” in *2021 European Control Conference (ECC)*, 2021, pp. 1577–1582. DOI: [10.23919/ECC54610.2021.9655166](https://doi.org/10.23919/ECC54610.2021.9655166)

Manuscripts and preprints

- [M1] S. Bodmer, D. Saccani, M. N. Zeilinger, and A. Carron, “Anytime plug-and-play control with contract-based distributed MPC,” Under review at *IEEE Transactions on Control Systems Technology*, 2026. arXiv: [2607.04215](https://arxiv.org/abs/2607.04215).
- [M2] J. Geurts, D. Saccani, M. N. Zeilinger, and A. Carron, “Distributed model predictive control with connectivity-based contracts,” Under review at *IEEE Robotics and Automation Letters*, 2026.
- [M3] L. Meroi, S. El Amrani, D. Saccani, D. Piga, and G. Ferrari-Trecate, “Risk-aware stability-preserving design of neural controllers with conformal verification,” Manuscript in preparation; intended for submission to *Automatica*, 2026.
- [M4] D. Saccani, “Spatial admissibility of secondary tasks in cooperative distributed MPC,” Under review at *IEEE Control Systems Letters*, 2026.

Book chapter

- [B1] D. Saccani, “Model predictive control for constrained navigation of autonomous vehicles,” in *Special Topics in Information Technology*, ser. SpringerBriefs in Applied Sciences and Technology, Springer, 2024, pp. 103–113. DOI: [10.1007/978-3-031-51500-2_9](https://doi.org/10.1007/978-3-031-51500-2_9)

Selected Talks, Workshops, and Academic Service

- Co-organizer and speaker, workshop *Distributed MPC for Multi-Agent Systems*, European Control Conference, Reykjavik, 2026.
- Co-organizer and speaker, workshop *Neural Network Control with Stability and Performance Guarantees*, European Control Conference, Stockholm, 2024.
- Invited talk, *Model Predictive Control for Constrained Navigation of Autonomous Vehicles*, IEEE ITSS Italian Chapter Annual Meeting, University of Genoa, 2024.
- Invited talk, *Multi-trajectory Model Predictive Control: Safe Trajectory Planning of Autonomous Vehicles*, EPF Lausanne, DECODE group, 2023.
- Reviewer for more than 50 submissions to international journals and conferences, including *Automatica*, *IEEE TAC*, *IEEE TCST*, *IEEE L-CSS*, *CDC*, *ECC*, and *ACC* (2020 – present).

Technical Skills and Languages

Methods: MPC, robust and distributed control, learning-based control, reinforcement learning, convex and nonlinear optimization. **Software:** Python, MATLAB/Simulink, ROS, PyTorch, CasADi, CVXPY. **Languages:** Italian (native), English (fluent).